

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

1st Semester of B. Pharm Examination

University Theory Examination Dec- 2016

MA131.1 Basic Mathematics

Date: 28/12/2016, Wednesday Time: 10:00 a.m. to 01:00 p.m. Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION I is 30 minutes. Answers to SECTION I must be written during the first 30 minutes of the examination.
3. Answers to SECTION II and SECTION III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat sketches wherever necessary.
6. Use of scientific non-programable calculator is allowed.

SECTION I

Q.1. Darken the circle of the correct/most suitable choice. [20]

1. A set of numbers $\{\dots, -2, -1, 0, 1, 2, 3, \dots\}$ is the set of

- (A) integers (C) nonnegative integers
(B) rational numbers (D) complex numbers

2. If $A = \{1, 3\}$ and $B = \{3, 4, 5\}$, $C = \{4, 5, 6, 7\}$ then $A \cup B =$

- (A) $\{1, 3\}$ (B) $\{3\}$ (C) $\{1, 3, 4, 5\}$ (D) $\{3, 4\}$

3. Suppose $n(U) = 70$, $n(A) = 30$, $n(B) = 45$ and $n(A \cup B) = 65$ then $n(A \cap B) =$

- (A) 5 (B) 25 (C) 10 (D) 20

4. Let f denote a function from A into B , $f : A \rightarrow B$ Then the set B is called

- (A) domain of f (B) range of f (C) subset of B (D) None of these

5. Consider $A = \{a, b, c\}$, $B = \{x, y, z\}$ then the relation $f = \{(a, x), (c, x), (a, z)\}$

- (A) is a set $A \times B$ (C) is a function $f : B \rightarrow A$
(B) a function $f : A \rightarrow B$ (D) is not a Function

6. Consider $A = \{a, b, c\}$, $B = \{1, 2, 3\}$ then the relation $f = \{(a, 2), (b, 1), (c, 2)\}$

is

- (A) not a one to-one function (C) not a function
 (B) onto function (D) a one-to-one function
7. If $y = \log_b x$ then (A) $b^y = x$ (B) $x^y = b$ (C) $b^x = y$ (D) $y^b = x$
8. Suppose $\lim_{x \rightarrow c} f(x) = 5$ and $\lim_{x \rightarrow c} g(x) = -2$, then $\lim_{x \rightarrow c} (f(x) + 3g(x)) =$
 (A) -1 (B) -6 (C) 8 (D) 3
9. A function $f(x)$ is continuous at an interior point $x = c$ of its domain if and only if
- (A) $f(c)$ exists (C) $\lim_{x \rightarrow c} f(x) = f(c)$
 (B) $\lim_{x \rightarrow c} f(x)$ exists (D) All three conditions
10. The slope of the curve $y = 1 - x^2$ at point $x = 2$ is
 (A) -3 (B) 4 (C) -4 (D) -2/3
11. If u and v are differentiable functions of x , then $\frac{d}{dx}(u + v) =$
 (A) $\frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$ (B) $\frac{du}{dx} + \frac{dv}{dx}$ (C) $v \frac{du}{dx} + u \frac{dv}{dx}$ (D) $v \frac{du}{dx} - u \frac{dv}{dx}$
12. The second derivative of the function $y = -x^2 + 3$ is
 (A) $-2x + 3$ (B) 3 (C) -2 (D) 0
13. An anti-derivative of the function $f(x) = 2x$ is
 (A) 2 (B) $\frac{x^2}{2}$ (C) $x^2 + C$ (D) $2 + C$
14. Let $\int_{-1}^1 f(x) dx = 5$, $\int_1^4 f(x) dx = -2$. Then, $\int_{-1}^4 f(x) dx =$
 (A) 7 (B) 5 (C) -7 (D) 3

15. The matrix $P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is

- Ⓐ diagonal Ⓒ lower triangular
Ⓑ upper triangular Ⓓ Identity

16. The matrix $F = \begin{bmatrix} 0 & -2 & -3 \\ 2 & 0 & -4 \\ 3 & 4 & 0 \end{bmatrix}$ is

- Ⓐ Symmetric Ⓒ lower triangular
Ⓑ upper triangular Ⓓ Skew- Symmetric

17. The inverse of matrix $\begin{bmatrix} -1/3 & 0 \\ 0 & 5 \end{bmatrix}$ is

- Ⓐ $\begin{bmatrix} -3 & 0 \\ 0 & 1/5 \end{bmatrix}$ Ⓑ $\begin{bmatrix} 3 & 0 \\ 0 & -1/5 \end{bmatrix}$ Ⓒ $\begin{bmatrix} -3 & 0 \\ 0 & -1/5 \end{bmatrix}$ Ⓓ $\begin{bmatrix} 3 & 0 \\ 0 & 1/5 \end{bmatrix}$

18. Which of the following matrices is singular?

- Ⓐ $\begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$ Ⓑ $\begin{bmatrix} 1 & 2 \\ 5 & 8 \end{bmatrix}$ Ⓒ $\begin{bmatrix} -1 & 6 \\ 3 & 9 \end{bmatrix}$ Ⓓ $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

19. The quadratic equation $ax^2 + bx + c = 0$ has two distinct roots if the value of $\Delta = \sqrt{b^2 - 4ac}$ is

- Ⓐ 0 Ⓑ grater than 0 Ⓒ less than 0 Ⓓ grater than 1

20. The median of numbers 45,36,28,42,45,40,44 is

- Ⓐ 42 Ⓑ 45 Ⓒ 43 Ⓓ 40

SECTION II

- Q.2.** Attempt any **Four** of the following. [20]
- A** Construct the venn diagram of the set relations [05]
 (a) $(A \cup B) \cap C'$ (b) $A \cap B \cap C'$ (c) $(A \cap B') \cup (A' \cap C)$ (d) $A \cap B \cap C$
- B** Consider the Universal set $S = \{1, 2, \dots, 9\}$ and the sets
 $A = \{1, 2, 3, 4\}, B = \{4, 5, 6, 7\}, C = \{5, 6, 7, 8, 9\}, D = \{1, 3, 5, 7, 9\}$ Find [05]
 (a) $A \cup B$ (b) $B \cup D$ (c) $A \cap (B \cap D)$ (d) $D \cup C'$
- C** 1. Sketch the graph of $f(x) = x^2 + x - 6$ [03]
 2. Let $A = \{1, 2\}, B = \{a, b, c\}$ and $C = \{c, d\}$ determine [02]
 (a) $(A \times C) \cup (A \times B)$ (b) $A \times (B \cap C)$
- D** Evaluate the limit $\lim_{x \rightarrow 4} \frac{4-x}{5-\sqrt{x^2+9}}$ [05]
- E** Find the first and second derivatives of the function $y = \frac{4x^3}{3} - x$ [05]
- F** Find the derivative of $y = \frac{\cos x}{1 - \sin x}$ [05]
- G** At what points is the functions $y = \frac{1}{x-2} - 3x$ continuous? [05]

SECTION III

- Q.3.** Attempt any **Four** of the following. [20]
- A** If $A = \begin{bmatrix} 2 & 3 & 1 \\ 6 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ Then verify $(AB)' = (B'A')$ [05]
- B** If $A = \begin{bmatrix} 2 & 3 & 1 \\ 6 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ Then verify $|AB| = |A||B|$ [05]
- C** 1. Evaluate $\int \left(\frac{\sqrt{x}}{2} + \frac{2}{\sqrt{x}} \right) dx$ [03]
 2. If $A = \begin{bmatrix} 2 & 3 \\ 1 & -4 \\ 1 & 0 \end{bmatrix}, B = \begin{bmatrix} -1 & 0 \\ 3 & 1 \\ 1 & -4 \end{bmatrix}$ and $C = \begin{bmatrix} 2 & 1 \\ 2 & 0 \\ 0 & 1 \end{bmatrix}$ Find $2A + B - 4C$ [02]
- D** Evaluate integral $\int_1^4 \sqrt{4x-1} dx$ [05]
- E** Find inverse of matrix A where $A = \begin{bmatrix} 1 & 0 & 0 \\ 5 & 4 & 0 \\ 6 & 2 & 5 \end{bmatrix}$ [05]
- F** Evaluate $\int (2x+1)^3 dx$ [05]

Q.4. Attempt any **Four** of the following. [20]

A 1. Solve the quadratic equation $3x^2 - 2x - 4 = 0$ [03]

2. Define arithmetic mean of variable. [02]

B Find the arithmetic mean of the following distribution: [05]

Table 1: measurements of glucose content (g per 100 mL) of orange juices

Glucose level	1.45 - 1.76	1.77 - 2.08	2.09 - 2.40	2.41 - 2.72	2.73 - 3.04
Number of samples	2	2	21	15	3

C The data in Table 2 shows raw calcium, milligrams per deciliter of 8 subjects 65 years of age and older. Compute

Table 2: laboratory values of elderly subjects

Subjects	1	2	3	4	5	6	7	8
CAMMOL	2.53	2.50	2.43	2.48	2.33	2.13	2.55	2.45

Arithmetic mean and standard deviation [05]

D The data in Table 3 shows the percentage of paracetamol in batches of Sterwin 500 mg tablets. The tablets were assayed by BP1993 UV method. Compute All quartiles [05]

Table 3: Paracetamol content (%) in batches of tablets

Batch No	1	2	3	4	5	6	7
UV	84.06	84.10	85.13	84.96	84.96	84.87	84.75

E Find median of the following distribution: [05]

Table 4: Frequency distribution of Age (in years)

Age	20-25	25-30	30-35	35-40	40-45	45-50
No. of Persons	170	110	80	45	40	35

F 1. Solve using Cramer's rule $x - 2y = 1$, $2x - 3y = 3$ [03]

2. Define median of variable. [02]